

AI-Powered 360 Panoramas: Unveiling Challenges for Realistic XR Prototyping

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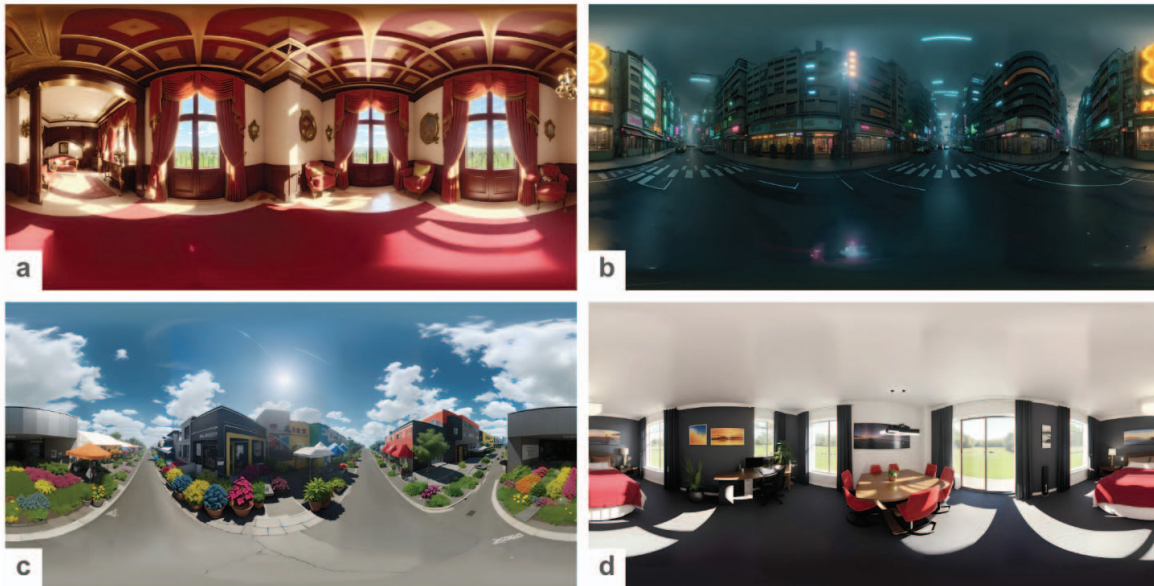


Figure 1: Different 360 panoramas generated based on text prompts using Skybox AI: a) living room¹; b) city at night²; c) market³; d) polished room⁴.

ABSTRACT

This paper explores and evaluates the main limitations associated with AI-based 360 panorama generation. We make use of a free AI-based 360 panorama generator, Skybox AI, to highlight some of the current limitations present in automatic panorama generation from text input. By recognizing and addressing these constraints, researchers and practitioners can pave the way for enhanced algorithms and systems capable of delivering more immersive and realistic panoramic experiences.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—Heuristic evaluations

1 INTRODUCTION

Augmented Reality (AR) / Virtual Reality (VR) development used to be time-consuming and required a lot of technical ability, how-

ever, with advancements in tools and techniques over the years, the process of turning ideas into prototypes has become more accessible [1, 2]. In recent years, the integration of Artificial Intelligence (AI) in the creation of Extended Reality (XR) content has marked a paradigm shift, offering novel avenues for immersive visual experiences [11]. Among the diverse applications of AI in XR, the generation of 360 panorama content stands out as a captivating area of exploration [10].

Unlike fully-rendered 3D scenes, 360 panoramas serve unique purposes within XR experiences. They offer immersive environments without the computational demands of real-time rendering, making them ideal for lower-powered devices and scenarios where real-time rendering isn't feasible. Additionally, 360 panoramas serve as background imagery for interactive experiences, such as skyboxes in VR environments, providing context and depth to virtual spaces.

We explored the state-of-the-art advancements in AI-driven 360 panorama generation [12], particularly emphasizing the extraction of spatial information from textual inputs. Leveraging the power of machine learning, these approaches hold the potential to revolutionize the way panoramic content is produced, enabling a seamless transition from descriptive text to immersive virtual environments.

As AI continues to evolve, its capacity to interpret and transform textual descriptions into visually rich XR environments also evolves. We analyzed AI techniques applied to 360 panorama generation, highlighting the strides made in understanding and translating textual input into compelling, spatially accurate panoramas. By elucidating the current achievements and challenges in this domain, we aim to contribute to the ongoing discourse surrounding the integration of AI in XR content creation, with a specific focus on 360 panoramas.

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¹<http://tinyurl.com/ENPTXRpano1>

²<http://tinyurl.com/ENPTXRpano2>

³<http://tinyurl.com/ENPTXRpano3>

⁴<http://tinyurl.com/ENPTXRpano4>

This paper is structured as follows. The Related work section provides some important works related to AI-based 360 panorama generation, together with some theoretical background. The Methodology section provides insight into our approach, combining a literature review of recent advancements with practical experimentation using an AI-based 360 panorama generation tool. The latter allows us to illustrate the tool’s capabilities and limitations through carefully designed scenarios. In sequence, the Scenario Design section details the textual prompts used to generate the 360 panoramas, showcasing the tool’s adaptability across diverse scenes. Notably, we highlight scenarios prone to exposing artifacts, shedding light on the tool’s performance under challenging conditions. Subsequently, our findings and insights regarding the limitations of the used tool are presented in the corresponding section. The Conclusion section emphasizes the identified limitations as crucial considerations for future developments in AI-driven panorama generation tools. We discuss the necessity for ongoing refinement and enhancement, bridging the gap between technical advancements and user-friendly accessibility.

2 RELATED WORK

Text-to-Image Generation has emerged as a significant research area bridging Natural Language Processing (NLP) and Computer Vision (CV). Leveraging advancements in Generative Adversarial Networks (GANs) [5], prior studies have concentrated on generating high-resolution images using stacked generators [14, 15], alongside enhancing image perceptual quality through the integration of regularization losses [3]. However, optimizing GANs poses challenges [6, 8] and encounters issues like mode collapse [9].

The main differences between AI-based image synthesis and AI-based 360 panorama synthesis, both originating from textual prompts, stem from the nature of the output and the complexity of the spatial context involved. In traditional image synthesis, the output is typically a 2D image that represents a scene or object described in the textual prompt. The focus is on generating a single static image that accurately reflects the described content. In contrast, 360 panorama synthesis involves generating a complete panoramic view that encompasses a full 360-degree field of view around a central point. The output is a spherical image that captures the surrounding environment from all angles, providing immersive spatial context.

Image synthesis from textual prompts often focuses on generating a single viewpoint or perspective of a scene or object described in the text. The spatial context is limited to the perspective captured within the 2D image frame. 360 panorama synthesis requires understanding and integrating spatial context from all directions around a central point. The AI system must consider not only the objects or scenes described in the text but also their placement and interactions within the entire panoramic environment.

While image synthesis from textual prompts presents its challenges, the complexity is often more manageable compared to 360 panorama synthesis. The AI system needs to understand and translate textual descriptions into coherent 2D images. 360 panorama synthesis introduces additional technical challenges, including spherical image mapping, seamless stitching of multiple viewpoints, and maintaining spatial consistency across the entire panoramic view. The AI system must effectively process and integrate spatial information to generate a realistic and cohesive 360-degree panorama.

A recent approach proposed by [43] introduces latent diffusion models for high-resolution image generation. This method encodes images into a low-dimensional latent space, perceptually equivalent to pixel space, thereby enhancing computational efficiency while preserving generation quality. Notably, diffusion models exhibit superior performance across various image generation tasks such as image inpainting [57], outpainting [54], and synthesis [38]. Furthermore, [46] extends diffusion models to 3D indoor scene synthesis, conditioning them on geometry and editing prompts, thus broaden-

ing their applicability to panoramic image generation tasks.

Wang et al. [13] proposed an approach that focuses on the customization of 360-degree panoramas, which inherently possess global geometric properties, using a T2I diffusion model. The fine-tuned model alone does not ensure the continuity between the leftmost and rightmost sides of the synthesized images, a crucial characteristic of 360-degree panoramas. By performing pre-denoising operations twice at each time step of the denoising process on the stitch block consisting of the leftmost and rightmost image regions, together with a global cropping method, it is possible to synthesize seamless 360-degree panoramas.

Chen et al. [4] proposed an approach for generating 360 panoramas from text prompts based on two steps. At first, an LDR (and low resolution) image is generated from the text prompt. Next, super-resolution inverse tone mapping is applied to scale up the LDR panorama both in resolution and dynamic range. The main limitation of their approach comprises the generation of 360 panoramas from terms that are not present in the training dataset. The generation only uses the terms that were present on the dataset, discarding the ones not present.

Li and Bansal [7] proposed an approach that goes beyond 360 panorama generation. They focused on navigation through 3D environments. This means that the panorama is regenerated according to the user navigation. They proposed a recursive image outpainting that reconstructs the missing observations in the panorama gradually based on text caption to generate coherent panorama with diverse objects and room layouts.

3 METHODOLOGY

To contextualize our study, we conducted a review of recent papers in the field of AI-driven 360 panorama generation. Notable contributions include Wang et al.’s work on “Customizing 360-Degree Panoramas through Text-to-Image Diffusion Models” [13], Chen et al.’s research on “ext2light: Zero-shot text-driven HDR panorama generation” [4], and Li and Bansal’s work on “PanoGen: Text-Conditioned Panoramic Environment Generation for Vision-and-Language Navigation” [7]. These studies laid the foundation for understanding the advancements and challenges in the field.

To illustrate the practical challenges faced by non-technical users, particularly designers, we opted for a user-friendly and freely available online tool called Skybox AI¹. Skybox AI allows users to generate 360 panorama content from textual descriptions without the need for programming skills. Our decision to use this tool was motivated by the desire to simulate a scenario wherein individuals without a technical background engage with AI-generated XR content creation.

Skybox AI presents a noteworthy option amidst the landscape of AI-driven 360 panorama generation tools. Its algorithmic efficiency and optimized workflows facilitate the swift creation of high-fidelity panoramas, effectively conserving computational resources and time for XR professionals. Demonstrating good accuracy and realism, Skybox AI produces immersive environments that enhance the perceptual fidelity of XR experiences. The tool’s intuitive user interface and accessible controls render it amenable to a broad spectrum of users, spanning from novices to seasoned practitioners.

We crafted a series of textual descriptions representing diverse scenes, including indoor and outdoor environments with varying levels of complexity. These descriptions were input into the Skybox AI platform, and the resulting 360 panoramas were analyzed for qualitative aspects.

To assess the performance of Skybox AI and highlight its limitations, we analyzed stitching quality, color consistency, and the ability to accurately represent spatial details.

Through a systematic exploration of the generated 360 panoramas, we aim to illustrate the tool’s limitations. This step contributes to

¹<https://skybox.blockadelabs.com/>

a nuanced understanding of the challenges that non-technical users might encounter in AI-driven 360 panorama generation tools like Skybox AI.

4 SCENARIO DESIGN

To systematically assess the performance of the Skybox AI tool, we designed a set of diverse textual descriptions representing various scenes. These descriptions aimed to cover a spectrum of indoor and outdoor environments, each posing unique challenges to the AI model. A total of four textual prompts were carefully crafted, considering factors such as scene complexity and spatial details. The selected textual descriptions include:

1. “a lavishly furnished living room with intricate details, such as patterned carpets, ornate furniture, and multiple light sources”: The complexity of indoor scenes with various textures, patterns, and intricate details can expose stitching artifacts and spatial distortions (Figure 1.a).
2. “a bustling city street at night, illuminated by a combination of neon signs, streetlights, and vehicle headlights”: The variation in lighting conditions and high contrast in night time city scenes can highlight color inconsistencies, tone mismatch, and challenges in handling diverse light sources (Figure 1.b).
3. “a dynamic scenario where the viewpoint rapidly changes, such as moving through a crowded market with stalls on either side”: Rapid changes in perspective can expose limitations in the tool’s ability to handle parallax effects and maintain spatial coherence during transitions (Figure 1.c).
4. “a room with large mirrors, polished surfaces, and abundant natural light streaming through windows”: Scenes with mirrors or reflective surfaces may challenge the stitching algorithm, leading to inaccuracies and distortions in the reflection of the surroundings (Figure 1.d).

The selected textual descriptions were input into the Skybox AI platform, and the resulting 360 panoramas were generated using the tool’s automated processes (the “Realism” style was selected for all generated panoramas). Skybox AI employs natural language processing algorithms to interpret textual input and translate it into a visually coherent XR environment. This method allowed us to simulate the user experience of non-technical individuals, particularly designers, engaging with the tool to create immersive panoramic content.

To explore the tool’s adaptability and limitations, we varied parameters such as scene complexity and level of detail in the textual prompts. By doing so, we aimed to observe how well Skybox AI could handle different scenarios and whether the generated panoramas accurately reflected the intended spatial and visual characteristics described in the text.

The evaluation criteria for the generated 360 panoramas encompassed key aspects such as stitching quality, color consistency, and the ability to faithfully represent spatial details. Additionally, we considered the tool’s responsiveness to nuanced textual input and its capacity to adapt to variations in scene descriptions. Some examples of the perceived artifacts are shown in Figures 2 to 5.

The scenario design process followed an iterative approach, refining the textual prompts based on initial outputs and user feedback. This iterative cycle allowed us to explore the tool’s learning capabilities and observe whether repeated use or nuanced prompts resulted in improved outcomes.

To complement the quantitative analysis, we sought feedback from a sample group of 3 computer science professors with more than 10 years of XR experience (2 male, 1 female, mean age: 44 years old) who interacted with the tool. They provided insights into

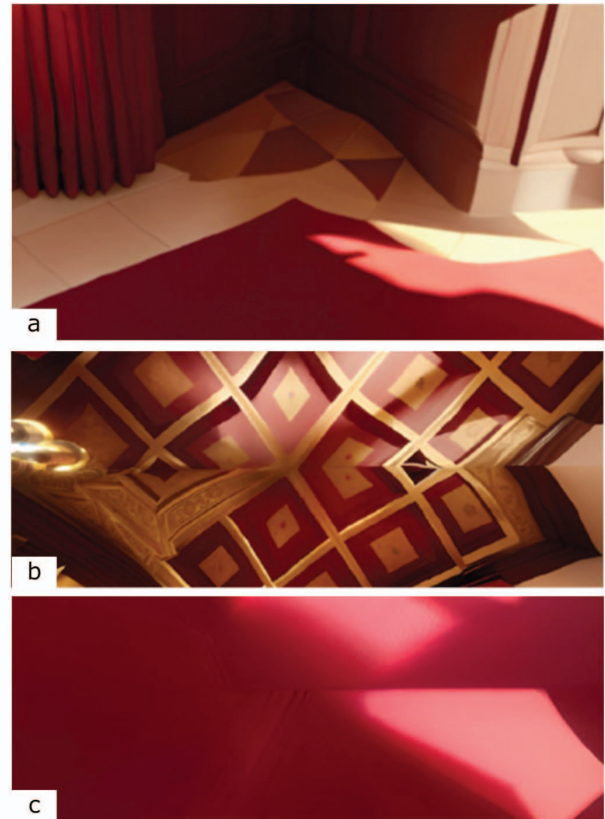


Figure 2: Some of the problems found on panorama #A: a) Spatial distortion found at a specific corner of the floor; b) Stitching/spatial distortion found at the top center; c) Stitching/color inconsistency found at the bottom center.

their user experience, highlighting aspects such as the tool’s intuitiveness, the degree of creative control, and challenges encountered during the panorama generation process.

The three professors were asked to use the tool with the four chosen text prompts, selecting no other option rather than the generated style (Realism). After generating the four required panoramas, they freely explored the tool’s other capabilities and user interface. The total time for the experiment was set to 20 minutes. All the visual problems found by them were grouped into a specific Appendix².

Through this carefully designed scenario approach, we aimed to comprehensively evaluate the capabilities and limitations of Skybox AI in generating 360 panoramas from textual descriptions, shedding light on its practical utility for non-technical users in XR content creation.

5 LIMITATIONS OF SKYBOX AI IN 360 PANORAMA GENERATION

The experimentation with Skybox AI revealed several noteworthy limitations that impact the tool’s efficacy, particularly when utilized by non-technical users such as designers. These limitations encompass technical challenges, perceptual inconsistencies, and user experience concerns:

- **Stitching artifacts and spatial distortions**

²<http://tinyurl.com/problemsfound>



Figure 3: Some of the problems found on panorama #B: a) Pattern generation problem (the pedestrian crossing showed random alignments) found at the bottom; b) Spatial misplacement (a light was placed floating in the middle of the air) found at the top.

The generated panoramas exhibited stitching artifacts, especially in scenes with complex spatial layouts or intricate details. Spatial distortions, such as misalignments and discontinuities, were evident in some instances. Such artifacts detract from the realism of the panoramas and may pose challenges for designers seeking high-fidelity representations of complex scenes.

- **Color inconsistency and tone mismatch**

Inconsistencies in color rendering and tone mismatch were observed across different scenes. This limitation was particularly evident in scenarios with diverse lighting conditions. Color inconsistencies can undermine the visual coherence of the generated panoramas, affecting the overall quality and believability of the XR content.

- **Ambiguity handling and interpretation challenges**

Skybox AI demonstrated difficulty in handling ambiguous or abstract textual prompts. Scenes with less explicit descriptions or metaphorical language often resulted in panoramas that deviated from the intended visual interpretation. Designers may face challenges when attempting to convey nuanced or abstract concepts through textual input, limiting the tool's adaptability to creative and varied scenarios.

- **Limited user control and customization**

The tool's interface offered limited options for users to exert control over specific aspects of the panorama generation process, such as adjusting lighting conditions, choosing focal points, or fine-tuning spatial elements. Designers seeking a higher degree of creative control may find the tool restrictive, hindering their ability to tailor panoramas to specific artistic or storytelling requirements.

- **Dependency on input quality**

The quality of the generated panoramas was highly dependent on the clarity and specificity of the textual input. Non-technical users may struggle to achieve desired outcomes if their textual prompts lack the precision required by the AI model, limiting the tool's accessibility and user-friendliness.



Figure 4: Some of the problems found on panorama #C: a) Spatial distortion found (the image shows the center of the street being mixed with the sidewalk) at the bottom; b) Color inconsistency found at the top.

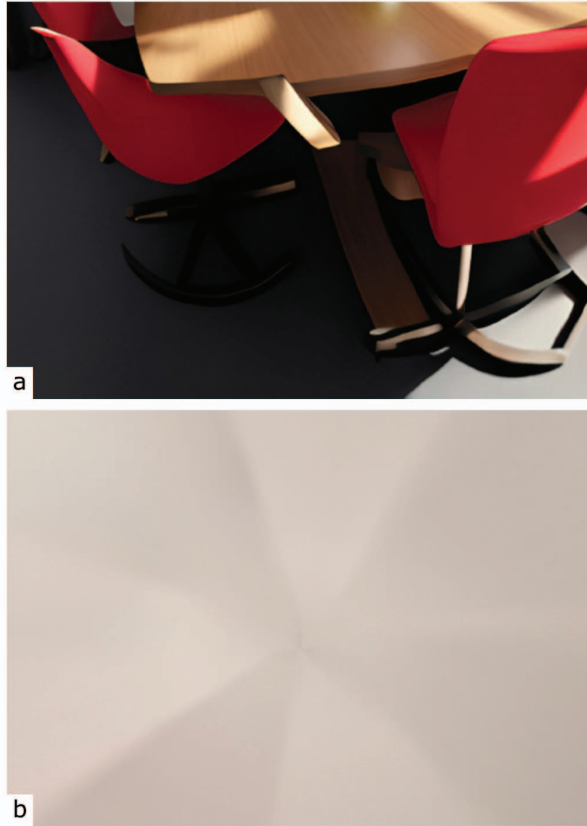


Figure 5: Some of the problems found on panorama #D: a) Inconsistent pattern at the table (also, one of the chairs seems incomplete); b) Stitching/color inconsistency problem found at the roof.

• Resource Intensiveness

The computational resources required for generating panoramas, especially in scenarios with high scene complexity, are significant. Since the panoramas are generated in the cloud, this aspect was not too perceivable by end users. Usually, it took from 20 to 25 seconds to generate each panorama.

6 CONCLUSION

The identified limitations highlight the need for continued refinement and enhancement in AI-driven 360 panorama generation tools like Skybox AI. While the tool offers a convenient entry point for non-technical users into XR content creation, addressing these limitations is crucial to unlock its full potential for diverse and nuanced panoramic storytelling.

Despite the identified limitations, it is important to recognize the growing significance of AI-based tools in the realm of panorama generation. Tools like Skybox AI serve as useful assets for non-technical users, offering a convenient entry point into the field of XR content creation. These AI-driven tools play a crucial role in expediting prototyping and validating projects, providing users with the capability to translate imaginative concepts into XR experiences.

Most issues encountered in the Skybox AI tool have already been addressed in recent research, such as in the study of Wang et al. [13]. However, a considerable number of state-of-the-art works are still disseminated through code repositories, often tied to a Google Colab environment. While this setup eases usability by developers, it remains distant from being accessible to non-technical users.

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